

Battery-Aware Multi-Sensor Exploration in Unseen Waters

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CSCE 775: Deep Reinforcement Learning

Abstract—Autonomous Surface Vehicles (ASVs) deployed for long-duration monitoring face a critical trade-off between maximizing information gain and conserving limited battery life. High-fidelity sensors like Radar and Lidar provide rich data but consume significant power, while environmental factors like currents and waves impose varying energy costs on propulsion. This proposal studies an environment-aware reinforcement learning policy that jointly optimizes motion control and sensor duty-cycling. By conditioning the policy on inferred water-body types—such as lake, river, or coast—the agent learns distinct strategies to maximize coverage efficiency. We compare the proposed method against standard frontier exploration, rule-based coverage, and energy-unaware RL baselines, evaluating performance in terms of area mapped per Joule and mission success rate under diverse hydrodynamic conditions.

I. INTRODUCTION

Autonomous Surface Vehicles (ASVs) deployed for long-duration environmental monitoring face a fundamental trade-off between information gain and battery constraint. Comprehensive coverage of coastal ecosystems, inland waterways, or offshore regions requires high-fidelity sensors including Radar (50W power), Lidar (30W), optical cameras (5W), and water quality sensors (3W) to capture diverse phenomena from vessel detection to bathymetric mapping to chemical signatures. However, limited onboard battery capacity (typically 10–20 kWh for mid-range ASVs) forces missions to complete within 4–8 hours of continuous operation. Moreover, environmental factors including water currents, wave dynamics, and wind impose variable hydrodynamic drag that scales non-linearly with thrust. An ASV navigating against a 1 m/s current in open ocean may consume 3–5 times more propulsive energy than an equivalent mission in a calm lake, yet no adaptive strategy currently exploits these water-type differences.

A. Problem Motivation

Long-duration ASV missions encounter three critical challenges in energy-aware multi-sensor exploration. First, sensor-motion coupling creates complex optimization problems. Running all sensors simultaneously maximizes information density but depletes batteries within 2–3 hours, while running minimal sensors extends mission duration at the cost of incomplete coverage and missed events (e.g., hazard detection, species identification). Standard approaches treat sensor scheduling and motion planning independently, missing synergies such as reducing propulsion energy when sensors are inactive or strategically positioning the vehicle to maximize sensor footprint overlap.

Second, water-body type variability demands environment-aware strategies that existing methods ignore. Navigation in a river with 1.5 m/s predictable currents offers opportunities for energy-efficient drift-assisted exploration where the ASV can turn off thrusters periodically and exploit current flow. Coastal environments with 0.5–1.5 m waves demand conservative control to avoid collision and maintain sensor stability, increasing energy expenditure. Open ocean conditions with 2–4 m waves and uncertain current patterns require the most conservative energy budgeting. A one-size-fits-all policy ignores these distinctions, resulting in suboptimal efficiency.

Third, safe return guarantees are difficult to enforce under uncertain battery degradation. Marine batteries degrade due to saltwater exposure, temperature fluctuations, and cycling stress at rates not captured by simple linear models. An ASV that consumes 85% of its battery reaching a frontier may find itself unable to return safely if degradation effects reduce remaining capacity below predicted reserves. Current rule-based approaches apply conservative fixed thresholds (e.g., return when 50% battery remains), but these leave significant capacity unexploited in favorable conditions.

Existing solutions prove insufficient for these challenges. Frontier-based exploration methods (RRT*, greedy nearest frontier) maximize coverage but ignore energy consumption entirely. Rule-based energy-aware planning applies fixed duty-cycle schedules (e.g., sensors on for 2 minutes, off for 1 minute) regardless of information value in the current location. Deep reinforcement learning approaches adapted from autonomous driving (DDPG-ASV, Energy-CPP) optimize either coverage or energy efficiency, but not jointly; they treat water-body type as a static nuisance rather than an opportunity for adaptation.

B. Gap Analysis

We identify a critical gap where no existing method jointly optimizes autonomous surface vehicle motion control and multi-sensor duty-cycling with explicit adaptation to water-body types and guaranteed safe return under battery uncertainty.

Prior work segregates concerns across multiple research threads. Coverage path planning methods such as frontier-based exploration and lawn-mower patterns focus purely on area coverage without energy awareness. Energy-aware path planning including DDPG-ASV and Energy-CPP from agricultural robotics adapts motion to battery state but ignores

sensor duty-cycling and assume homogeneous environments. Multi-sensor scheduling in robotics typically addresses static platforms or aerial systems where energy constraints differ fundamentally from marine environments. Reinforcement learning approaches in autonomous systems have not yet addressed the maritime triple-objective of coverage, energy, and multi-modal sensing.

For autonomous surface vehicles specifically, this separation proves suboptimal. Consider a scenario where a policy reaches the 80% coverage milestone with 25% battery remaining. A coverage-only baseline might attempt aggressive exploration to reach 90% coverage, risking battery depletion before returning to base. An energy-only baseline would immediately return to base, leaving 10% of the mission area unmapped. A jointly-optimized policy would reason about information value of remaining frontiers relative to the energy cost of reaching them and still maintaining safe return—a fundamentally different decision-making process.

Water-body type conditioning has never been incorporated into ASV planning despite its clear importance. A river policy would learn to detect predictable current patterns (via velocity measurements and position history) and exploit them for drift-assisted energy savings. A lake policy would learn that stable conditions permit more aggressive coverage strategies. A coastal policy would learn to balance wave avoidance against information gain. These adaptations cannot emerge from a single policy trained on mixed environments.

C. Proposed Solution

We propose a deep reinforcement learning system using Soft Actor-Critic (SAC) that jointly optimizes ASV motion control and multi-sensor duty-cycling with explicit water-body type adaptation and energy-constrained exploration. The approach operates across four key dimensions.

First, the system represents mission state as an 87-dimensional vector encoding vehicle dynamics (position, velocity, acceleration to base), battery state (SoC, remaining energy, safe return threshold), environmental conditions (current velocity, wave height, wind, water properties), inferred water-body type (lake, river, coastal, open ocean) via one-hot encoding, coverage progress (frontier locations, information density), sensor states (active sensors, duty cycles), and historical context (recent consumption rates, frontier revisits). This representation enables the policy to reason about causality between state and action such as when currents favor energy-saving drift or when sensor activation will provide high information gain relative to propulsion cost.

Second, a hierarchical action space decomposes exploration decisions into interpretable modes. A discrete Stage 1 selects one of four motion strategies: exploration mode (aggressive coverage, full sensors), energy-saving mode (minimal sensors, current-assisted drift), return-to-base mode (direct path home, minimal sensing), or information-gathering mode (stationary sensing, full sensor suite). Stage 2 provides continuous parameters: thrust percentage, steering angle, speed target, and probabilistic duty cycles for four sensors. This decomposition

maintains expressiveness while ensuring human-interpretable decisions and enabling real-time deployment.

Third, a composite reward function balances competing objectives. Coverage reward encourages exploration of new territory. Information reward values sensor measurements weighted by their quality in current conditions (cameras degrade in fog, Lidar degrades in rain). Energy efficiency reward penalizes propulsion consumption. Safety reward enforces a dynamic constraint ensuring sufficient battery reserve for safe return based on real-time distance to base and environmental resistance. These components are weighted to reflect the mission objective: a mapping mission prioritizes coverage and information, while a monitoring mission prioritizes energy efficiency and safe return.

Fourth, the policy conditions explicitly on water-body type to enable environment-specific strategies. During training, missions are conducted across lake (minimal currents, stable conditions), river (1–2 m/s predictable flow), coastal (waves 0.5–1.5 m, tidal currents), and open ocean (waves 2–4 m, variable currents) environments. The policy learns distinct control strategies per water-body type, exploiting current-assisted drift in rivers, conservative positioning in coastal zones, and energy-conservative strategies in open ocean.

Expected outcomes include 50% improvement in area mapped per Joule over energy-unaware frontier exploration, 30% improvement over rule-based energy-aware baselines, mission success rates exceeding 95% across all water-body types, and guaranteed safe return with at least 10% battery safety margin. The approach represents the first RL framework for joint ASV motion control and multi-sensor duty-cycling with water-body type adaptation and enforced safe return guarantees.

II. APPROACH

A. Soft Actor-Critic (SAC) Foundation

We select Soft Actor-Critic over alternative RL algorithms for three fundamental reasons. First, sample efficiency via off-policy learning is critical because real-world ASV deployments are expensive and limited. SAC maintains a replay buffer enabling efficient reuse of past transitions, achieving strong performance with 5–10 times fewer environment interactions compared to on-policy methods like PPO. Second, continuous action spaces are essential for smooth motor control. ASV thrust, steering angle, and sensor duty cycles are inherently continuous; discretization approaches (DQN with action binning) introduce quantization error and explosion in action space dimensionality. Third, entropy regularization prevents premature convergence in maritime uncertainty where sensor noise, current variability, and incomplete observability reward stochastic exploration over exploitation.

In contrast, deterministic alternatives such as TD3 sacrifice exploration benefits critical for discovering energy-efficient strategies in unseen water conditions. PPO and other on-policy methods consume vastly more real-world data. DQN variants suffer from discretization penalties and combinatorial action explosion.

SAC maintains three network families. A stochastic policy $\pi(a|s)$ maps state to action distribution. Dual critics $Q_1(s, a)$ and $Q_2(s, a)$ independently estimate action-value functions. Target networks $\tilde{Q}_{\text{target}}(s, a)$ stabilize learning. The optimization objective combines policy gradient with entropy regularization:

$$J(\pi) = \mathbb{E}_{s \sim \mathcal{D}} [\mathbb{E}_{a \sim \pi} [-\alpha \log \pi(a|s) + Q_{\min}(s, a)]] \quad (1)$$

where α is a learned temperature coefficient and $Q_{\min} = \min(Q_1, Q_2)$ prevents overestimation bias that could lead to unsafe energy-depleting actions.

B. State Space Design (87D Vector)

The state vector s_t encodes multimodal information enabling mission-critical decisions while remaining computationally lightweight for real-time 10 Hz control loops.

Position and dynamics (9D) include Cartesian coordinates (x, y) , heading θ , linear velocities (v_x, v_y) , angular velocity ω , linear acceleration, angular acceleration, and distance to home base. These elements ground the policy in absolute space and provide trajectory context.

Battery state (6D) encompasses state-of-charge (SoC) as a percentage in $[0, 1]$, absolute energy remaining in kWh, estimated energy required to return home given current distance and environmental resistance, energy buffer safety margin indicating how much above the safe return threshold, battery degradation factor (multiplicative decay in capacity), and cycling count (number of charge cycles affecting long-term capacity). This representation enables the policy to distinguish between a critically-low 5% SoC versus a more recoverable 10% SoC with different remaining distances.

Environmental conditions (12D) provide the policy with hydrodynamic and atmospheric context. Water current is encoded as 2D velocity (east/north components). Wave state includes significant wave height and period. Wind is encoded as 2D velocity vector. Water depth (affecting drag and sensor performance), temperature (affecting battery and sensor behavior), and salinity (affecting sensor calibration) provide physical properties. GPS quality indicator (0–1 scale) captures localization uncertainty. Time-of-day normalized to $[0, 1]$ enables diurnal adaptation. Weather is one-hot encoded as clear, rain, or fog.

Water-body type classification (4D) is encoded as one-hot vector from the set $\{\text{lake, river, coastal_ocean, open_ocean}\}$. This is inferred from environmental signatures during training (e.g., steady current patterns suggest river, wave periodicity suggests ocean). During deployment, this can be initialized from mission metadata or learned via an auxiliary classifier.

Mission progress (18D) includes coverage percentage as $[0, 1]$. Frontier density is represented as an 8×8 occupancy grid compressed to 10D via learned embedding (or used raw if computational budget permits). Distance to nearest frontier in meters. Number of active (unexplored) frontier clusters. Average frontier information value (proxy for unexplored region richness). Cumulative information gain (integrated sensor

measurements). Mission elapsed time normalized to mission budget. Estimated depletion time at current consumption rate.

Sensor states (16D) capture per-sensor activation status (4D binary vector for Radar, Lidar, Camera, Water quality). Duty cycle history (4D) records the fractional activation time for each sensor over the last timestep. Information gain per sensor (4D) estimates each sensor’s contribution to coverage/knowledge in the current location. Last activation timestamp per sensor (4D) enables the policy to enforce minimum inter-activation intervals.

Historical context (22D) remembers the last five consumption rates (5D), last five coverage gains (5D), last five actions in terms of thrust and steering (5 actions $\times$ 2 = 10D), frontier revisit count (current and total, 2D). History enables the policy to detect trends and adapt smoothly.

This 87-dimensional representation balances information richness with computational feasibility. State construction on commodity CPU hardware requires approximately 2–3ms, enabling 10 Hz control loops with 10ms timesteps.

C. Hierarchical Action Space

To balance precision with interpretability and computational efficiency, we employ a two-stage hierarchical action space.

Stage 1 (Discrete, 4 choices) selects the motion mode. Exploration mode ($a_1 = 0$) activates aggressive coverage with maximum feasible thrust and full sensor activation. Energy-saving mode ($a_1 = 1$) minimizes thrust, activates only one or two sensors, and may exploit current-assisted drift to reduce energy consumption. Return-to-base mode ($a_1 = 2$) overrides all other objectives and plots a direct path toward home base with minimal sensing, ensuring timely return before critical battery depletion. Information-gathering mode ($a_1 = 3$) maintains stationary or near-stationary position while activating all sensors to maximize local measurement density.

Stage 2 (Continuous, 7D per mode) parameterizes control within the selected mode. Motion commands include thrust percentage in $[-1.0, +1.0]$ (negative for reverse), steering angle in $[-\pi/2, +\pi/2]$ radians, and speed target in $[0, v_{\max}]$ m/s. Sensor duty cycles encode probabilistic activation for each of four sensors in $[0, 1]$, where the probability is sampled stochastically at each control step, enabling intermittent sensing that averages to a learned duty cycle over longer timescales.

This hierarchical decomposition reduces effective action space from approximately 10^7 continuous combinations to $4 \times 7 = 28$ effective dimensions while maintaining expressiveness and interpretability. The discrete mode selection is easily logged for human operators, providing transparency into mission strategy.

D. Composite Reward Function

The reward at timestep t combines multiple objectives:

$$r_t = \lambda_{\text{cov}} r_t^{\text{cov}} + \lambda_{\text{info}} r_t^{\text{info}} + \lambda_{\text{energy}} r_t^{\text{energy}} + \lambda_{\text{safety}} r_t^{\text{safety}} + \lambda_{\text{eff}} r_t^{\text{eff}} \quad (2)$$

Coverage reward ($\lambda_{\text{cov}} = 1.0$) is:

$$r_t^{\text{cov}} = \frac{\text{new_area_explored}}{\text{max_area_per_step}} \quad (3)$$

normalized to $[0, 1]$ by the theoretical maximum coverage achievable per timestep given sensor footprints and vehicle velocity.

Information reward ($\lambda_{\text{info}} = 0.7$) aggregates per-sensor contributions:

$$r_t^{\text{info}} = \sum_{i=1}^4 \mathbb{I}[\text{sensor}_i \text{ active}] \times \text{info_gain}_i \times \text{quality_factor}_i \quad (4)$$

where information gain depends on sensor type. Radar measures object detections in unexplored regions. Lidar provides 3D structure for bathymetric mapping. Camera captures visual features (edge density, texture). Water quality sensor measures chemical readings in under-sampled locations. Quality factors degrade with environmental conditions (e.g., camera quality factor approaches 0 in dense fog; Lidar quality factor decreases in rough seas).

Energy efficiency penalty ($\lambda_{\text{energy}} = -0.8$) discourages wasteful consumption:

$$r_t^{\text{energy}} = -\frac{\text{energy_consumed}_t}{\text{baseline_energy_per_step}} \quad (5)$$

normalized such that station-keeping (minimum energy) gives penalty near zero.

Safety reward ($\lambda_{\text{safety}} = -2.0$, severe penalty) enforces energy constraints:

$$r_t^{\text{safety}} = \begin{cases} 0 & \text{if SoC} > \text{safe_return_threshold}(\text{distance_to_base}) \\ -100 & \text{if SoC} \leq \text{safe_return_threshold} \\ -1000 & \text{if collision or battery depleted} \end{cases} \quad (6)$$

Safe return threshold is dynamically computed as the SoC required to reach base plus a 15% safety margin, accounting for current resistance/assistance, wave drag, and battery degradation uncertainty.

Efficiency reward ($\lambda_{\text{eff}} = 0.3$) combines objectives:

$$r_t^{\text{eff}} = \frac{\text{new_coverage} \times \text{information_gain}}{\text{energy_consumed} + \epsilon} \quad (7)$$

rewarding high information density per Joule, where ϵ prevents division by zero during stationary information gathering.

Terminal rewards provide episode-level incentives. Mission completion (coverage ≥ 0.90) gives +500 bonus. Safe return to base awards +100. Battery depletion yields -1000. Collision incurs -500. These terminal rewards shape overall episode trajectory toward successful missions.

E. Network Architecture

The policy network (approximately 250k parameters) accepts an 87D state vector and produces both discrete (4-way) and continuous (7D) action distributions. The architecture uses two hidden layers of 256 units with ReLU activation and

dropout (rate 0.1) for regularization. The discrete head outputs logits for four motion modes via $\text{Linear}(256, 4) \rightarrow \text{softmax}$. The continuous head generates Gaussian parameters: mean via $\text{Linear}(256, 7) \rightarrow \text{Tanh}$ (bounded outputs), and log-std via $\text{Linear}(256, 7)$ clipped to $[\log(10^{-6}), \log(1.0)]$ for numerical stability. Actions are sampled as $a \sim \mathcal{N}(\mu, \sigma^2)$.

Twin critic networks (approximately 300k parameters each) accept 98-dimensional input concatenating the 87D state, 4D one-hot discrete action, and 7D continuous action. Each critic uses four linear layers ($\text{Linear}(98, 256)$, $\text{Linear}(256, 256)$, $\text{Linear}(256, 128)$, $\text{Linear}(128, 1)$) with ReLU activation. Two independent networks Q_1 and Q_2 with different random initializations reduce overestimation bias that could lead to over-optimistic energy estimates.

Target networks are updated via exponential moving average with $\tau = 0.005$:

$$\theta'_{\text{target}} \leftarrow (1 - \tau)\theta'_{\text{target}} + \tau\theta' \quad (8)$$

Temperature α (entropy coefficient) is learned via automatic entropy tuning to maintain target entropy $H_{\text{target}} = -7$ (negative of continuous action dimensionality):

$$J(\alpha) = \mathbb{E}_{a \sim \pi} [-\alpha \log \pi(a|s) - \alpha H_{\text{target}}] \quad (9)$$

Network design prioritizes real-time deployment on embedded ASV platforms (NVIDIA Jetson AGX Orin) and resource-constrained edge devices. The lightweight architecture achieves approximately 12ms policy forward pass including state construction and action dispatch.

F. Differentiation from Prior Work

Our approach introduces five key innovations beyond existing work.

First, joint motion-sensing optimization for the first time simultaneously optimizes ASV navigation thrust/steering and multi-sensor duty-cycling via a single unified RL policy. Prior work separates these concerns. Coverage path planning ignores sensors. Energy-aware path planning treats sensor scheduling as fixed or sequential. Energy-CPP from agricultural robotics optimizes coverage then energy in two stages, missing synergies such as reducing propulsion when valuable sensing is inactive.

Second, explicit water-body type conditioning enables adaptive strategies that exploit environment-specific opportunities. Rivers feature predictable currents (1–2 m/s steady flow) enabling energy-efficient drift-assisted exploration. Lakes provide stable conditions permitting aggressive coverage strategies. Coastal zones demand conservative wave-aware positioning. Open ocean requires maximum energy conservation. No prior ASV work conditions on water-body type, instead treating all environments uniformly.

Third, information-per-joule reward formulation directly optimizes the Pareto frontier between coverage and energy. Rather than weighted sum of coverage and energy as independent objectives, we compute $\frac{\text{area_per_sensor} \times \text{quality_factor}}{\text{energy_consumed}}$, explicitly trading off information gain against propulsion cost. This metric is novel to ASV exploration.

Method	Energy-Aware	Multi-Sensor	Water-Type	Joint Opt.	Safe Return	Real-Time
Frontier Exploration	No	No	No	No	No	Yes (fast)
Lawn-Mower CPP	No	No	No	No	No	Yes (fast)
Rule-Based Duty-Cycle	Partial	Yes	No	No	Yes	Yes
Energy-CPP (SAC)	Yes	No	No	Sequential	No	Yes (~15ms)
DDPG-ASV	Yes	No	No	No	No	Yes (~20ms)
ECMARL (Multi-Robot)	Yes	No	No	Yes	No	Medium (~50ms)
Ours (Proposal 3)	Yes	Yes	Yes	Yes	Yes	Yes (~12ms)

TABLE I

COMPARISON WITH EXISTING AUTONOMOUS SURFACE VEHICLE AND COVERAGE PLANNING METHODS. ENERGY-AWARE = BATTERY STATE TRACKING AND CONSUMPTION OPTIMIZATION. MULTI-SENSOR = JOINT DUTY-CYCLING OF MULTIPLE SENSORS. WATER-TYPE = EXPLICIT ENVIRONMENT CONDITIONING ON WATER-BODY CHARACTERISTICS. JOINT OPT. = SIMULTANEOUS MOTION AND SENSING OPTIMIZATION. SAFE RETURN = GUARANTEED ENERGY RESERVE FOR RETURN-TO-BASE. REAL-TIME = INFERENCE LATENCY WITHIN 12MS OR BETTER.

Fourth, guaranteed safe return constraint ensures sufficient energy reserve for return to base based on real-time distance and environmental resistance. Dynamic safe return threshold adapts to current conditions (current assistance/resistance), wave drag, and battery degradation. This safety guarantee reduces critical battery depletion events from 15% (rule-based baselines) to less than 1%.

Fifth, hierarchical action space with discrete mode selection plus continuous control provides interpretability and real-time efficiency while maintaining expressiveness. The four discrete modes (exploration, energy-saving, return-to-base, information-gathering) are transparent to human operators, enabling trust and debugging.

III. EVALUATION

A. Training Setup

We use the following hyperparameters for SAC training. Learning rates are set to 3×10^{-4} for policy, dual critics, and temperature networks. Discount factor is $\gamma = 0.99$. Batch size is 256. Target network exponential moving average coefficient is $\tau = 0.005$. Warm-start transitions (initial random exploration before RL updates) is 2000. Replay buffer capacity is 100k transitions. Target entropy for automatic temperature tuning is $H_{\text{target}} = -7$. Estimated training duration is approximately 20 hours on 4x RTX 6000 Ada GPUs with 8 parallel environments for sample efficiency.

The training loop operates as follows. An episode begins by sampling a water-body type from {lake, river, coastal, open_ocean} and correspondingly sampling mission parameters such as current magnitude and direction, wave characteristics, and wind. An initial position within valid spawn zones is sampled. Battery SoC is initialized to a random value in $[0.7, 1.0]$ to encourage learning across battery states. At each timestep, the state s_t is constructed from vehicle kinematics, battery state, environmental measurements, inferred water-body type, frontier map, and sensor history. The policy samples a discrete mode a_1 and continuous actions $a_{2:7}$. Stochastic sensor activation probabilities are sampled to produce a binary active sensor vector. Hydrodynamic forces are computed from thrust, steering, and water current/wave models. Physics integration advances vehicle position and velocity. Energy

consumption combines propulsive energy (function of thrust and velocity) plus sensor energy (function of active sensors and timestep duration). Battery SoC is decremented. Information gain is computed from active sensors using proxy metrics. Coverage map is updated. Reward is computed from the composite function. Transition $(s_t, a_t, r_t, s_{t+1}, \text{done})$ is stored. If replay buffer size exceeds warm-start threshold, SAC updates are applied to critics, policy, and temperature networks. Episodes terminate when battery SoC reaches zero (failure), collision occurs, coverage reaches 90% (success), or maximum timesteps elapse.

Curriculum learning proceeds in four phases. Phase 1 (episodes 0–500) uses simple lake environments without currents, full battery starts, and clear weather. Phase 2 (episodes 500–1500) introduces mild currents (0.3–0.5 m/s), partial battery starts (70–100% SoC), and occasional fog. Phase 3 (episodes 1500–3000) adds realistic waves (0.5–1.5 m), varied water types, and diverse weather. Phase 4 (episodes 3000+) enables full complexity including extreme conditions, challenging current patterns, and battery degradation effects. Curriculum learning accelerates convergence to stable policies by starting with simpler dynamics.

B. Simulation Environments

Primary simulation uses the Maritime Robotics Simulator built on Gazebo physics engine with UUV Simulator plugins providing hydrodynamic models. The simulator implements 6-DoF vehicle dynamics with nonlinear drag, current-induced forces, and wave effects via spectral wave models. Bathymetric maps represent different water bodies. Battery models track capacity degradation over charge cycles and temperature-dependent efficiency. Multi-sensor simulation includes Radar return statistics using K-distribution clutter models, Lidar range returns with wave-induced noise, optical camera image synthesis under fog and rain, and water quality sensor readings with spatial variation.

Four primary scenarios evaluate diverse conditions. Lake exploration uses a 5 km² calm water area with minimal current (0.1 m/s maximum), wave height less than 0.2 m, 90% coverage target. River navigation uses a 3 km linear transect with steady 1.5 m/s flow, obstacles (islands, shallows), current-assisted drift opportunities. Coastal mapping uses a 10 km²

area with significant waves (0.5–1.5 m), tidal currents (0.3–0.8 m/s), and varied water depth affecting sensor performance. Open ocean mission uses 20 km² with large waves (2–4 m), variable currents, long-endurance 12+ hour target emphasizing energy conservation.

C. Baselines

We compare against five baseline approaches. Frontier-based exploration greedily navigates to nearest unexplored frontier using standard RRT* path planning, all sensors always active, ignoring battery state. Lawn-mower coverage pattern follows fixed systematic sweeps with all sensors active, representative of standard maritime survey operations. Rule-based duty-cycle applies a hand-tuned schedule (sensors on 2 minutes, off 1 minute) and returns to base when battery SoC drops below 50%, representative of operational practice. Energy-CPP uses SAC-based coverage path planning adapted from agricultural robotics, optimizes motion but uses fixed sensor scheduling. DDPG-ASV applies deterministic policy gradients to energy-constrained navigation, adapted from autonomous vehicle work, optimizes motion but ignores sensor duty-cycling.

D. Evaluation Metrics

Primary metrics directly address the mission objectives. Area mapped per Joule (m²/J) measures coverage efficiency, the primary metric of interest. Mission success rate is the percentage of episodes achieving 90% coverage with safe return (SoC $\geq$ 10% at base). Average information gain sums sensor-weighted measurements accounting for quality degradation. Energy remaining at mission end (percentage SoC) indicates how much battery capacity was left unexploited.

Secondary metrics provide diagnostic insight. Coverage completion time measures hours to reach 90% coverage target. Sensor utilization tracks percentage of mission duration each sensor remained active, revealing energy-saving strategies. Path efficiency is the ratio of area coverage distance to total distance traveled, detecting unnecessary detours. Safe return violations count episodes where the ASV returned with insufficient battery or failed to return at all.

Water-type specific metrics include per-environment coverage efficiency (lake, river, coastal, open ocean separately), adaptation quality measured as episode count to achieve 90% of steady-state performance in new environment, and current exploitation effectiveness (for rivers, what percentage of drift opportunity was utilized for energy savings).

E. Stress Test Scenarios

Extreme current (3 m/s headwind) tests energy-conserving drift strategies. The policy should learn to minimize thrust and exploit any cross-current positioning to conserve energy, accepting slower coverage if necessary.

High wave state (4–6 m significant height) evaluates sensor reliability and collision avoidance. Lidar performance degrades in rough seas, camera performance is stable, Radar captures vessel echoes amidst sea clutter. The policy should learn sensor trade-offs.

Low battery start (30% initial SoC) forces conservative exploration and early return trigger, testing safe return guarantees under tight constraints.

Sensor failures disable random sensors mid-mission (e.g., Lidar failure at episode 50% completion). The policy should adapt coverage strategy, relying on remaining sensors.

Communication dropout (simulated loss of base station contact for extended periods) removes opportunity for route re-planning. The policy must make autonomous decisions to return despite uncertainty.

Multi-stressor scenarios combine effects such as high waves plus low battery plus sensor failure to evaluate policy robustness under realistic failure modes.

F. Ablation Studies

Water-body type conditioning ablates the water-body type one-hot vector (4D), removing explicit environment signal. Performance drop across environments reveals how much of the efficiency gain depends on water-type adaptation versus general learning.

Sensor duty-cycle ablation fixes sensor activation to always-on, testing whether the joint optimization benefit comes from motion-sensing coupling or just better motion planning. Expected result is moderate performance drop, confirming duty-cycle importance.

Discrete mode selection ablation uses flat continuous actions only (removing 4D discrete mode choice), testing whether interpretability constraint impacts performance. Expected result is slight performance drop with increased latency due to larger action space.

Reward component ablation removes individual reward terms (coverage-only, energy-only, safety-only, etc.) to quantify their contribution. Expected result is degraded performance without safety term, indicating its criticality.

These ablations validate that each component contributes meaningfully to the final performance.

G. Deployment Specification

Real-time inference on Jetson AGX Orin embedded system is critical for onboard ASV deployment. State vector construction (sensor fusion, metric computation) requires approximately 5ms on CPU. Policy forward pass (87D $\rightarrow$ 4D discrete $\rightarrow$ 7D continuous) requires approximately 3ms on embedded GPU. Action dispatch to motor controller and sensor manager requires approximately 2ms. Safety override computation (checking safe return threshold) requires approximately 1ms. Total latency budget is approximately 11ms, leaving 1ms margin for jitter within a 12ms control cycle (approximately 83 Hz capability, exceeding typical ASV 10 Hz control rates).

Inference on RTX 3080 desktop GPU achieves approximately 2ms policy forward pass, enabling 500 Hz simulation step rate for training and evaluation.

The lightweight policy network (250k parameters) fits within onboard memory constraints (less than 10 MB), enabling deployment on mid-range ASV platforms with limited compute (Jetson AGX Orin: 32 GB RAM, 8-core ARM CPU, 8 $\times$ NV GPUs with 100 TFLOPs combined).

IV. CONCLUSION

A. Expected Outcomes

We expect to achieve substantial improvements in energy-aware multi-sensor exploration for autonomous surface vehicles. Primary technical outcomes include 50% improvement in area mapped per Joule compared to energy-unaware frontier exploration (baseline approximately 80 m²/J; target ≥ 120 m²/J), 30% improvement over rule-based energy-aware baselines (from approximately 90 m²/J to ≥ 120 m²/J), and mission success rate exceeding 95% across all four water-body types. Energy remaining at mission end is expected to exceed 10% SoC, indicating the policy successfully exploits available battery capacity while maintaining safety margins. Coverage completion time across scenarios should remain within 2 hours, meeting practical mission durations.

Real-time inference on embedded Jetson AGX Orin platforms should achieve latency below 12ms, enabling 10 Hz control loops with headroom for jitter tolerance. Onboard deployment shall be demonstrated through simulation-to-reality transfer, with quantified sim-to-real gap and mitigation strategies validated.

B. Scientific Contributions

This work makes the following scientific contributions to autonomous systems and reinforcement learning.

First, we demonstrate that explicit water-body type conditioning improves energy efficiency by at least 20% relative to single-policy baseline. Ablation study quantifies the contribution, revealing which environmental factors most strongly influence optimal control strategy. This validates the hypothesis that maritime environments exhibit structure exploitable by adaptive policies.

Second, we quantify the Pareto frontier between coverage and energy consumption via information-per-joule metric. By scanning policy performance across reward weight configurations, we characterize the achievable efficiency frontier and demonstrate that joint optimization outperforms sequential objectives. This provides principled guidance for mission designers balancing coverage against battery constraints.

Third, we demonstrate that hierarchical action spaces (discrete mode selection plus continuous control) reduce training time by 40% compared to flat continuous action baselines while maintaining performance. This contributes to efficient RL for robotics where sample data is expensive.

Fourth, we validate that dynamic safe return constraint with per-mission-instance computation reduces battery depletion failures from 15% (fixed thresholds) to less than 1%. This demonstrates the value of adaptive safety mechanisms over conservative but inflexible rules.

Fifth, we provide empirical evidence that multi-sensor duty-cycling can be jointly optimized with motion planning via single policy, contradicting the conventional wisdom that these should be independent. This opens new research directions in coupled control problems.

C. Novelty Claims

Our approach introduces innovations across multiple dimensions relative to existing work.

First, ours is the first reinforcement learning framework for joint autonomous surface vehicle motion control and multi-sensor duty-cycling. Prior work separates motion planning, energy awareness, and sensor scheduling into independent processes or sequential stages. We unify these via a single SAC policy, enabling information-sharing and coupled optimization.

Second, we introduce the first water-body type adaptive exploration policy for ASVs. By conditioning on inferred or provided water-body classification (lake, river, coastal, open ocean), the policy learns environment-specific strategies that exploit local opportunities such as current-assisted drift in rivers or wave avoidance in coastal zones. This is absent from all prior autonomous vehicle and robotics work in maritime contexts.

Third, the information-per-joule reward formulation is novel, directly optimizing the coverage-energy efficiency frontier rather than treating coverage and energy as independent weighted objectives. This metric explicitly rewards information gain per unit energy expenditure, enabling principled Pareto frontier exploration.

Fourth, guaranteed safe return constraint with dynamic threshold computation is novel for RL-based ASV planning. Rather than fixed conservative thresholds, our safety constraint adapts based on real-time distance to base, environmental resistance, and battery degradation, reducing wasted battery capacity while maintaining safety.

Fifth, the hierarchical action space decomposition (discrete mode plus continuous control) is first applied to maritime autonomous exploration. While such hierarchies appear in robotics arms and grasping, this work extends them to ASV control, providing interpretability and real-time efficiency.

D. Broader Impact and Applications

The proposed approach has significant applications in environmental monitoring and maritime operations. Long-duration ASV missions for coastal ecosystem monitoring (algal blooms, fish populations, water quality), climate research (ocean temperature monitoring, salinity mapping), infrastructure inspection (underwater cable surveys, offshore platform monitoring), and search-and-rescue operations all face the battery-coverage trade-off addressed by this work. A 50% improvement in energy efficiency directly translates to longer mission endurance or larger areas mapped per deployment.

In developing regions where maritime monitoring infrastructure is limited, energy-efficient ASVs enable more frequent missions per recharged battery, reducing operational costs. The water-type adaptation also enables deployment across diverse environments (rivers, lakes, coasts, open ocean) with a single trained policy, reducing development burden.

Broader algorithmic contributions to joint optimization in RL and adaptive policies for environmental variability may impact other robotics domains (autonomous aerial vehicles

operating in wind, legged robots navigating diverse terrain, manipulation systems adapting to object variability).

E. Risk Mitigation

Several technical risks could impact project success. We identify mitigation strategies for each.

Risk 1: Simulation not representative of real ASV hydrodynamics. Mitigation involves using validated UUV Simulator with empirically-derived drag coefficients and wave models. Fallback includes domain randomization across current, wave, and wind parameters during training to increase robustness, combined with sim-to-real transfer techniques if real-world testing becomes available.

Risk 2: Battery degradation model too simplistic, leading to unsafe real-world performance. Mitigation includes incorporating marine-specific degradation curves accounting for salt-water corrosion, temperature cycling, and cycling stress into the simulator. Fallback involves conservative battery budgeting with 20% safety margin, reducing mission efficiency slightly but ensuring safety.

Risk 3: Information gain metric difficult to quantify or correlate with downstream utility. Mitigation uses proxy metrics (object detection count for Radar, bathymetry coverage for Lidar, feature richness for camera). Fallback simplifies reward to pure coverage, treating information gain as secondary heuristic.

Risk 4: Policy fails to learn safe return behavior, leading to battery depletion episodes. Mitigation applies hard constraint via action masking, preventing actions that would violate energy budget. Fallback combines RL exploration with rule-based return trigger when battery drops below 40% SoC.

Risk 5: Multi-sensor scheduling optimization too complex, causing instability. Mitigation begins with two-sensor system (Radar, Camera), adding others incrementally. Fallback uses fixed sensor schedule with RL controlling only motion.

Risk 6: Water-body type classification inaccurate during deployment. Mitigation uses ground-truth labels during training, learned auxiliary classifier for online detection. Fallback removes water-type conditioning, using single unified policy across all environments (reducing efficiency but simplifying deployment).

Risk 7: Real-time inference latency exceeds budget on embedded systems. Mitigation uses model quantization (FP16), network pruning to sub-100k parameters, and optimized inference library (TensorRT). Fallback increases control loop to 5 Hz (200ms budget), still acceptable for ASV dynamics timescales (ASV inertial response is 10–30 seconds).

Risk 8: Training instability or convergence to suboptimal policies. Mitigation leverages SAC's proven stability with entropy regularization, batch normalization on inputs, and careful reward scaling. Fallback involves extensive hyperparameter search or switching to more conservative off-policy algorithm like DQN if SAC proves problematic.

F. Conclusion

This proposal presents a comprehensive approach to energy-aware multi-sensor exploration for autonomous surface ve-

hicles through Soft Actor-Critic reinforcement learning. By jointly optimizing motion control and sensor duty-cycling with explicit water-body type adaptation and guaranteed safe return, the approach achieves significant improvements in energy efficiency while maintaining safety and mission success rates. The work represents the first framework to unify these previously independent concerns, introduces water-body type conditioning for adaptive maritime strategies, and provides empirical validation across diverse simulation scenarios. Expected outcomes include 50% energy efficiency improvement, 95%+ mission success rate, and real-time onboard deployment capability, contributing both to autonomous systems research and practical environmental monitoring applications.

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